Academic Success Modeled by Socioeconomic Predictors

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I. Abstract

Education is a basic human right, and higher education is linked to better outcomes later in life. ACT scores are one of the qualifications that are reviewed for access to higher learning institutions. But what are the socioeconomic factors that may influence academic success, and therefore educational sucess? This report modeled ACT scores based on various socioeconomic predictors based with a multiple linear regression model and compared to actual average ACT scores reported by high schools across the United States. Among several predictors, percent of the population receiving free or reduced lunch was the strongest predictor of ACT scores, followed by census tract unemployment rate and percent of parents with a college degree. Further study should seek to understand how schools with free or reduced lunch programs prioritize college preparedness.

II. Introduction

Education is considered to be a basic human right. It has a powerful ability to lift children and adults out of poverty (UNESCO, 2022). Education level is a widely recognized indicator of future success; in the United States, the unemployment rate of college graduates and is consistently lower than the working population graduation rate ((U.S Bureau of Labor Statistics, 2026). A student's entrance to higher education is based on several factors, one of which is performance on college preparedness exams such as the ACT or SAT. This project aims to examine the socioeconomic indicators that contribute to academic achievement. Academic achievement will be measured by an individual high school's reported average ACT score. EdGap provides an average ACT score or SAT equivalent for each high school in the dataset. It is important to note that not every state requires students to take the ACT or SAT, which can lead to a greater proportion of high-performing students taking the exams (EdGAP, 2016). Additionally, the National Center for Education Statistics provides basic identifying information about the schools. More detailed socioeconomic characteristics by state are provided by the US Census Bureau. The report will aim to determine the most poignant indicators that predict academic success.

III. Theoretical Background

This project will use a various regression models to predict ACT scores based on given indicators. Both a simple linear regression model and a quadradic regression model will be simulated and assessed for best fit. The model with the best fit will have both a high R-squared value, which signals the strength of the variable's relationship to the data, and a small p-value, which signals that the probability of the correlation is high.

IV. Methodology

The School Information data set provides various information about the schools, including the charter type, grading types, and more. The selected school information is school year, state, zip code, level, and charter. The selected columns from the EdGap data are school ID, unemployment rate, adults with college degree, median household income, percent of married parents, ACT average, and percent free or reduced lunch. The School Information and EdGap data frames were merged with the school ID as the key. The US Census dataset provided a plethora of employment and income related information; however, this analysis focused on percent of population receiving food stamps or SNAP benefits, and percent of population without health care. Those two metrics were selected to attempt to highlight regions where many students may face food insecurity and limited access to healthcare, which could result in lower academic performance. The census information was joined using the state as a key. Any socioeconomic variables that were out of range or illogical were set to NaN. For example, the percentage of students who are receiving a free lunch cannot be a negative number, and the average ACT score cannot report a value less than 1. Any observations that were missing the average ACT score were omitted. Missing socioeconomic data was estimated using an iterative imputer from sklearn.

After some preliminary data analysis, the median income variable was assessed for its ability to predict ACT score. First, average ACT score and median income were fitted with a simple linear regression model, and the RSME and MAE values were calculated. Then this was repeated with a quadratic linear regression model. The two models were then compared using an analysis of variance to determine if the difference between the two is statistically significant.

Next, the following variables were used in a multiple linear regression model: rate of unemployment, percent of parents with college education, percent of married parents, median income, percent receiving free lunch, percent receiving food stamps/SNAP, percent without health insurance. The top three indicators were selected and used for another multiple linear regression model. The two models were then compared to determine if there is a statistically significant difference between the full model and the reduced model Finally, all indicators were normalized to zero with a standard deviation of one. The normalized regression model was created to determine the relative influence of each predictor.

V. Computational Results

Correlation Coefficients - Normalized Predictors

Predictor	Correlation Coefficient	Standard Error	P-Value
Unemployment Rate	-2.0881	0.213	0.000
Percent of Parents with College Education	1.7804	0.158	0.000
Percent of Married Parents	-0.0560	0.134	0.676
Median Income	-1.741e-07	1.22e-06	0.888
Percent Receiving Free Lunch	-7.6637	0.100	0.000
Percent Receiving Food Stamps/SNAP	0.0062	0.012	0.612
Percent without Health Insurance	0.0205	0.005	0.000

VI. Results

From the analysis, it is clear the median income is a weak predictor of ACT score. Though it had a small root mean square error and mean absolute error which supports that there is a correlation. Transformations of the median income did not result in a significant change in correlation. Figure 1 shows the relationship between the median income and average ACT score. While the quadratic linear regression model provided a closer representation, it is still a weak correlation. Interestingly, there appears to be a much higher variation in ACT scores in school districts with lower median income than the high-income school districts.

Of the seven selected socioeconomic indicators, percent of the students receiving free lunch had the highest correlation coefficient, indicating the strongest indicator of ACT score. Though the correlation coefficient is negative, indicating an inverse relationship. The unemployment rate was the next highest predictor of ACT score, followed by percent of parents with college education. The former has an inverse relationship whereas the latter has a positive relationship. The two indicators from the US Census Economic data had low correlation coefficients: Percent receiving food stamps/SNAP and percent without health insurance.

The multiple linear regression model with the seven indicators was able to predict average ACT score with a high R-squared value, indicating a good fit. However, the reduced

model with only the top three predators was able to predict ACT score with similar levels of accuracy. The R-squared values for the two were the same to the hundredth's places.

VII. Conclusions

Percent of students receiving free lunch and unemployment rate are the two predictors with the strongest negative relationship with academic success, as measured by ACT score. Whereas percent of parents with a college degree has the highest positive relationship with average ACT score. Other factors such as married parents, receiving food stamps, median income, and access to healthcare showed a weak influence on academic success. The regression model created with only the top three indicators did not have a substantially different correlation.

The Census data was limited in its influence, likely due to the data being reported at the state level. Within one state, there is often substantial variability between urban areas versus rural areas. Future studies should attempt to explore similar topics at a finer level. The percent of the population without health care was a weaker predictor of ACT scores, though it was not the lowest of the seven. It shows promise to have a greater relationship with data at the census tract level.

This report highlights an interesting detail about the inverse relationship between free lunch programs and ACT scores. Future study should attempt to uncover why schools that have a high percentage of students in free lunch programs have an inverse relationship with academic success. Are these schools more or less likely to have additional college preparatory resources? This report offers promising directions for future research.

VIII. Appendix

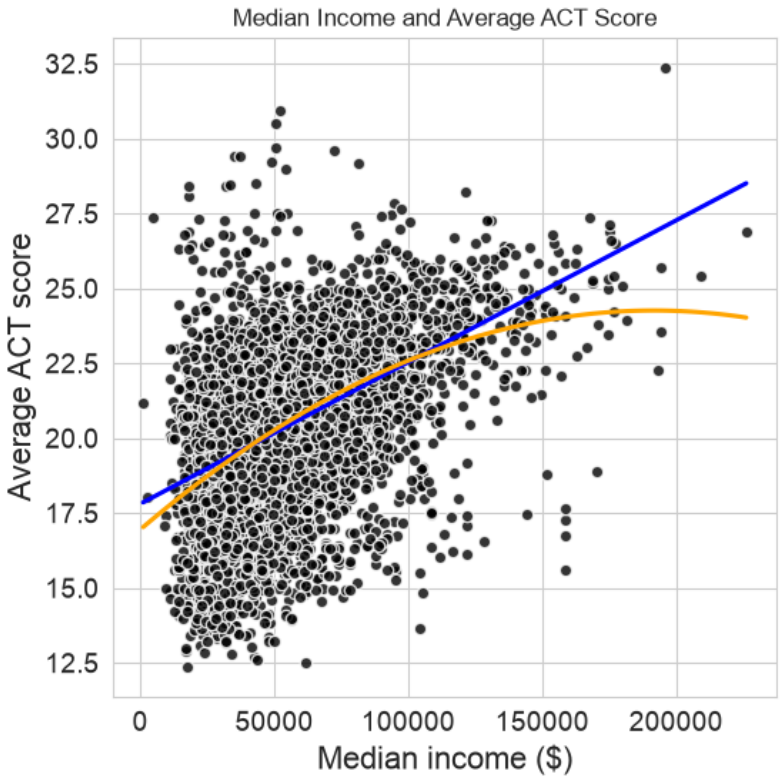


Figure 1: Shows the scatterplot of average ACT score and median income with the linear regression model (blue) and quadratic linear regression model (orange).

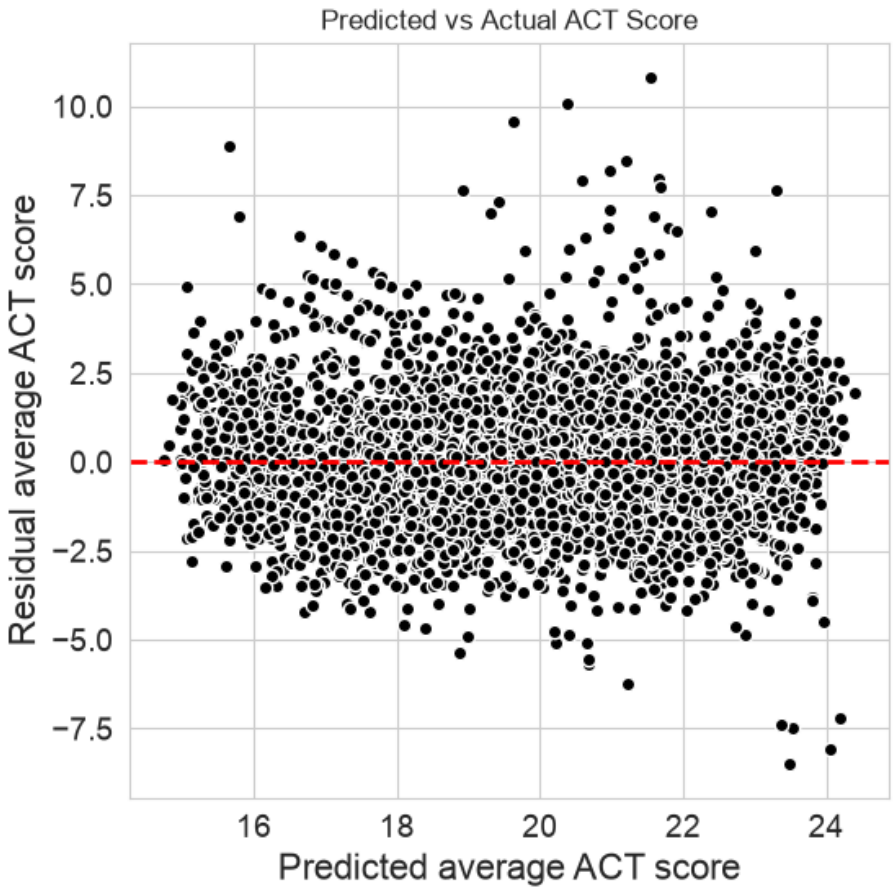


Figure 2: Scatterplot shows the predicted ACT score based on the full linear regression model and the actual average ACT score.

IX. References

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